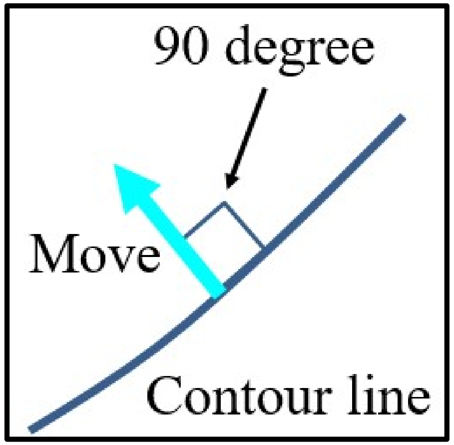
Gradient Descent (Algorithm: Constant Step Size)

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha \nabla f(\mathbf{x}^{(k)})$$

α : Step size (positive constant).



Lab Session Task 1

Choose an initial solution in [-5, 5] x [-5, 5], specify the step size, and show a sequence of moves from the initial solution using the gradient decent algorithm with the constant step size for the following problem (see below). By iterating these steps using a different initial solution and a different step size, show several sequences of moves. Then, choose the most appropriate step size for this problem. That is, your task is to clearly show the search behavior and to find the best specification of the constant step size.

Minimize $f(\mathbf{x}) = (x_1 + 1)^2 / 9 + (x_2 + 1)^2$

Lab Session Task 2

Do the same as Task 1 for the following problem:

$$\text{Minimize} \quad f(\boldsymbol{x}) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$